

Attention to the Macroeconomy

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August 2025

CEBRA 2025 – Boston

Motivation

- Attention is key to belief formation and decision-making.
 - Attention to the economy is central for business cycles, and policy transmission.
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 - Attention increases belief accuracy, rationally traded-off against attention costs.
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- **Behavioral economics:** Attention may not be “goal-optimal”.
 - Selective attention, salience, memory, information avoidance, ...
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 - Selective attention, salience, memory, information avoidance, ...
(Bordalo et al., 2025; Hartzmark et al., 2021; Sicherman et al., 2016)
- Empirical properties of attention to the economy and link to beliefs not well understood due to a lack of direct data on attention allocation.

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- New panel data measuring attention to the macroeconomy.
 - Large quarterly surveys of German households and firms between December 2020 and March 2023, i.e., before and during the post-pandemic shock to inflation.

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- Theoretical benchmarks from a model of goal-optimal attention.
- Empirical facts about attention, beliefs and link to actions.
 - Purely descriptive, but naturally occurring variation in attention, large samples, period with a changing economic environment.
 - We link attention and beliefs to decisions (firms' price setting behavior).

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 - Purely descriptive, but naturally occurring variation in attention, large samples, period with a changing economic environment.
 - We link attention and beliefs to decisions (firms' price setting behavior).
- Develop [model of selective memory recall](#) to reconcile theory and empirics.
 - Test additional predictions in the data.

Preview of results

- **Consistent with theories of goal-optimal attention:**
 - Persistent cross-sectional heterogeneity in attention, related to exposure & info costs.
 - Attention higher when environment is more volatile.
 - Attention associated with more frequent expectation adjustment, higher confidence, smaller misperceptions of realized inflation.
 - Firms attentive to inflation are more likely to plan or to have implemented price increases.

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- **At odds with goal-optimal attention:**
 - Attentive agents deviate more strongly upward in their inflation expectations from expert benchmarks.
 - Disagreement about nowcasts is lower than about forecasts among attentive agents.

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- **At odds with goal-optimal attention:**
 - Attentive agents deviate more strongly upward in their inflation expectations from expert benchmarks.
 - Disagreement about nowcasts is lower than about forecasts among attentive agents.
- **Extended model with similarity-based memory recall:**
 - Reconciles theory and empirics.
 - Model predicts **important role for experiences** in agents' attention allocation and belief formation: **confirmed in our data.**

Related literature

- **Determinants of information acquisition and attention allocation to the economy.**

(Adam, 2007; Bracha and Tang, 2024; Capozza et al., 2022; Coibion and Gorodnichenko, 2015; Coibion et al., 2018; Flynn and Sastry, 2024; Goldstein, 2023; Korenok et al., 2023; Mikosch et al., 2024; Pfäuti, 2024a,b; Roth et al., 2022; Song and Stern, 2024; Weber et al., 2025)

↪ We provide **direct field evidence** on the dynamics, consequences, and drivers of attention to the macroeconomy.

- **Role of memory and experiences in belief formation.**

(Afrouzi et al., 2023; Andre et al., 2022; Bordalo et al., 2024, 2023a,b, 2020; Conlon and Patel, 2025; D'Acunto and Weber, 2022; Enke et al., 2024; Gennaioli et al., 2024; Gödker et al., 2021; Goldfayn-Frank and Wohlfart, 2020; Graeber et al., 2024; Kahana, 2012; Laudenbach et al., 2024; Malmendier and Nagel, 2016; Salle et al., 2023)

↪ Our results point to a role for experiences – through **similarity-based recall** – in shaping the link from attention to beliefs.

Outline

- ① Setting and Data
- ② A Model of Goal-Optimal Attention
- ③ Testing Theories of Goal-Optimal Attention
- ④ Selective Memory and Non-Goal-Optimal Attention

Setting and Data

Context and timeline

- We conduct **quarterly surveys** with German households and firms b/w December 2020 and March 2023 (10 waves).
 - Firms: ifo Business Survey ($N \approx 3,000$ per wave).
 - Households: Dynata, broadly representative of German population ($N \approx 5,000$ per wave).

▶ Summary statistics

▶ Responses/Attrition

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[▶ Summary statistics](#) [▶ Responses/Attrition](#)
- Turbulent economic times:
 - Recovery from the pandemic recession.
 - Surge in inflation (largely unexpected) starting mid-2021 to close to 10% end-2022, amplified by Ukraine war and energy crisis. [▶ More](#)
 - Interest rate at the ZLB until mid-2022, then sharp rate hikes up until 3.5% in March 2023.
 - Unemployment rate relatively stable between 5% and 6%.

Open-ended measurement of attention allocation

- **Attention as the allocation of cognitive resources across (economic) domains.**
- We want to be agnostic about the exact margins through which attention matters:
 - Information acquisition.
(Mankiw and Reis, 2002; Reis, 2006a,b)
 - Information processing.
(Maćkowiak and Wiederholt, 2009; Sims, 2003; Woodford, 2003)
- We rely on an **open-ended elicitation**:
 - Increasingly common in economics.
(Andre et al., 2022, 2025, 2024; Haaland et al., 2025; Stantcheva, 2021, 2023)
 - Core advantage compared to more structured question formats: **no priming** through displayed response options.

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Q: *What topics come to mind when you think about the economic situation of your company/household?*

Coding of open-ended responses

Coding scheme for responses to the open-ended question to quantitatively analyze the unstructured text data:

- Codes for a range of topics:
 - **Macroeconomic:** pandemic, inflation, growth, monetary policy, ...
 - **Household-level:** income, spending, saving, housing costs, ...
 - **Firm-level:** supply chain, input factors, product demand, costs, ...
- Each response can be assigned multiple codes.
- Instruct research assistants to apply the coding scheme to the open-text responses.

Reliability of coding scheme & validity of open-ended data

- 85.5% of HH responses and 97.3% of firm responses can be assigned at least one code.
 - Double-coding of subset: High inter-rater reliability ($\approx 90\%$).
 - Large overlap with automated word counts. [▶ More](#)
 - Large overlaps with AI-based classification. [▶ More](#)
- Coding scheme **reliably captures content** of open-ended data.

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→ **Coding scheme reliably captures content of open-ended data.**

- Strongly correlated w/ attention measure based on structured survey question. [▶ More](#)
- Attentive agents consume more economic news. [▶ More](#)
- Time variation closely aligned with Google searches. [▶ More](#)

→ **Supports validity of open-ended measurement.**

A Model of Goal-Optimal Attention

Model setup

Standard model of rational inattention (Maćkowiak et al., 2023):

- deviating from the full-information forecast leads to losses ...
- ...but acquiring and processing information is costly.

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Setup:

- (demeaned) Inflation follows an AR(1)-process with persistence ρ_π .
- Loss is given by:

$$\frac{B}{2} \left(\underbrace{\rho_\pi \pi}_{\text{full info}} - \pi^e \right)^2$$

- Cost of information is linear in mutual information: $C = \kappa I(\pi; s)$.
- Gaussian prior: $\pi \sim N(\hat{\pi}, \sigma_\pi^2)$, we set prior mean $\hat{\pi} = 0$ for simplicity.

Model solution

- Optimal attention choice:

$$\gamma_{\pi} = \max \left\{ 0, 1 - \frac{\kappa}{B\rho_{\pi}^2\sigma_{\pi}^2} \right\},$$

where B is the agent's exposure to inflation and κ is the cost of information.

- Inflation nowcast: $\tilde{\pi} = \gamma_{\pi}s$.
- Inflation forecast: $\pi^e = \rho_{\pi}\gamma_{\pi}s$.

Model predictions

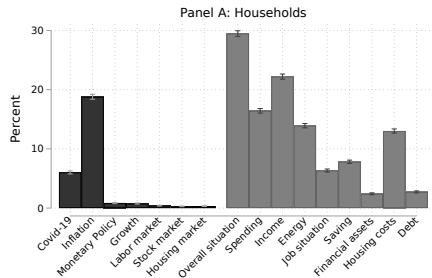
- **Variation in attention:** agents pay more attention if ...
 - they are more exposed to inflation.
 - they face lower information costs.
 - the variable to be forecast is more volatile (higher prior uncertainty).

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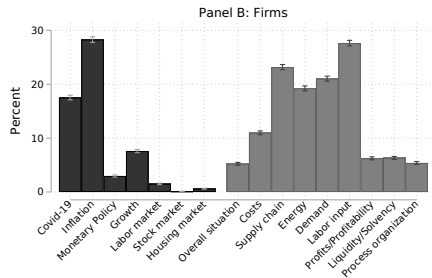
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 - they are more exposed to inflation.
 - they face lower information costs.
 - the variable to be forecast is more volatile (higher prior uncertainty).
- **Attention and beliefs:** **attentive agents** ...
 - are more likely to **adjust their expectations** when obtaining a new signal.
 - are **more confident** in their beliefs.
 - make **more accurate nowcasts and forecasts**:
 - expected **nowcast error** $|\pi - \gamma_\pi \pi|$ is lower with γ_π closer to 1.
 - expected **forecast error** $|\rho_\pi| \cdot |\pi - \gamma_\pi \pi|$ is lower with γ_π closer to 1.
 - **Attention is goal-optimal**
 - disagreement about nowcasts $|\gamma_{\pi,1}s_1 - \gamma_{\pi,2}s_2|$ is (weakly) larger than about forecasts $|\rho_\pi| |\gamma_{\pi,1}s_1 - \gamma_{\pi,2}s_2|$.

Testing Theories of Goal-Optimal Attention

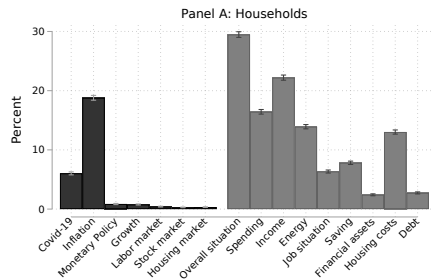
Variation of attention across topics



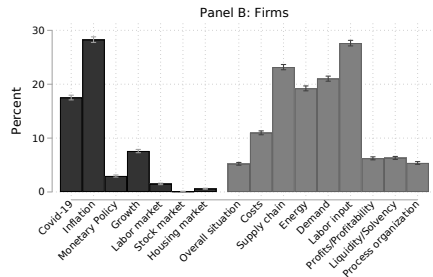
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- Attention to the macroeconomy concentrated in inflation.



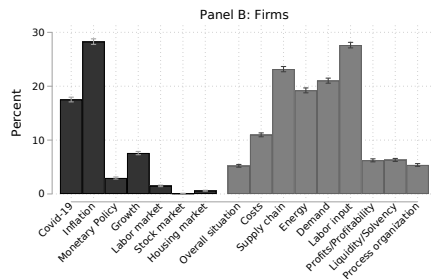
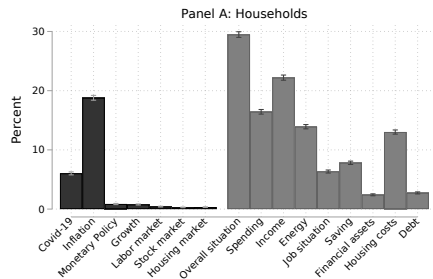
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- **Individual fixed effects** are an important source of variation. [▶ More](#)
- **Time fixed effects** explain a meaningful but smaller part of the variation. [▶ More](#)



Variation of attention across topics



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- Time fixed effects explain a meaningful but smaller part of the variation. [▶ More](#)
- Dynamics: attention shifts from Covid-19 to inflation, consistent with increasing inflation volatility and changes in news coverage. [▶ More](#)

Cross-sectional variation in attention: Measures of attention drivers

- **Direct measure of perceived exposure to different variables:**
 - Self-reported on a 5-point categorical scale.
- **Information acquisition costs** (only available for households):
 - Perceived difficulty of finding macroeconomic information.
 - Self-reported on a 5-point categorical scale.

⇒ regress individual fixed effects in attention on these measures (+ demographics)

Cross-sectional variation in attention: Households

	Attention			
	(1) Inflation	(2) Monetary policy	(3) Growth	(4) Any macro topic
Low information acquisition costs	0.030*** (0.007)	0.003* (0.002)	-0.001 (0.002)	0.046*** (0.009)
High perceived exposure to variable	0.097*** (0.005)	0.005*** (0.001)	0.007*** (0.001)	0.126*** (0.007)
Demographic controls	Yes	Yes	Yes	Yes
Observations	10,755	10,755	10,755	10,755

- Consistent with the model predictions, [attention](#) ...
 - [decreases with information costs](#).
 - [increases with proxies for exposure](#) to the variable to be forecast.

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Next: [how is attention to inflation related to beliefs about inflation?](#)

Attention and beliefs about inflation

	Absolute change in ex- pectation ≥ 0.5 p.p.	Confidence (z)	Expected inflation	Absolute deviation from expert forecast	Perceived current inflation	Absolute deviation from current level
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: Households						
Attention to inflation	0.021*** (0.007)	0.167*** (0.016)	0.171** (0.087)	0.104 (0.085)	-0.111 (0.072)	-0.502*** (0.061)
Distinct respondents	6,716	10,755	10,755	10,755	8,328	8,328
Observations	20,983	34,976	34,976	34,976	24,404	24,404
R-squared	0.02	0.12	0.16	0.10	0.14	0.07
Mean dep. var.	0.79	0.04	7.08	4.88	6.32	2.67
SD dep. var.	0.41	0.99	6.50	6.17	5.26	4.26
Panel B: Firms						
Attention to inflation	0.013** (0.006)	0.044** (0.017)	0.212*** (0.046)	0.200*** (0.046)		
Distinct respondents	4,402	6,193	6,235	6,235		
Observations	18,423	27,121	28,107	28,107		
R-squared	0.02	0.02	0.49	0.23		
Mean dep. var.	0.80	0.04	5.47	3.00		
SD dep. var.	0.40	1.02	3.44	2.72		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

- Attentive agents **update inflation expectations more frequently** in our environment of changing signals, consistent with the model.

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- Attentive agents **more confident** in their expectations, consistent with the model.

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- Smaller misperceptions about current inflation, consistent with the model.

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- Expectations deviate more strongly (upward) from experts, at odds with model.
- Forecast disagreement *exceeds* disagreement in nowcasts, at odds with model.

► More

Firms' price plans and attention to inflation ► Definition

	Planned price change (trichotomous)		Planned price increase (dummy)		Planned price decrease (dummy)	
	(1)	(2)	(3)	(4)	(5)	(6)
Attention to inflation	0.101*** (0.009)	0.086*** (0.009)	0.095*** (0.008)	0.078*** (0.008)	-0.007** (0.003)	-0.008*** (0.003)
Distinct respondents	6,178	4,873	6,178	4,873	6,178	4,873
Observations	28,198	26,893	28,198	26,893	28,198	26,893
R-squared	0.10	0.44	0.12	0.45	0.03	0.37
Mean dep. var.	0.42	0.42	0.46	0.46	0.04	0.04
SD dep. var.	0.57	0.57	0.50	0.50	0.20	0.19
Controls	Yes	No	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes	No	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

- More attentive firms are more likely to plan price increases.

Realized price changes and attention to inflation

► Definition

	Realized price change last month (trichotomous)		Realized price increase last month (dummy)		Realized price decrease last month (dummy)	
	(1)	(2)	(3)	(4)	(5)	(6)
Attention to inflation	0.063*** (0.008)	0.059*** (0.008)	0.057*** (0.007)	0.055*** (0.007)	-0.006** (0.003)	-0.005 (0.003)
Distinct respondents	6,184	4,878	6,184	4,878	6,184	4,878
Observations	28,366	27,060	28,366	27,060	28,366	27,060
R-squared	0.10	0.43	0.13	0.45	0.03	0.41
Mean dep. var.	0.25	0.25	0.29	0.29	0.04	0.04
SD dep. var.	0.51	0.51	0.45	0.45	0.19	0.19
Controls	Yes	No	Yes	No	Yes	No
Firm FE	No	Yes	No	Yes	No	Yes
Time FE	Yes	Yes	Yes	Yes	Yes	Yes

- More attentive firms are more likely to have implemented price increases.
- Attention correlated with firms' decision-making.

Selective Memory Recall and Non-Goal Optimal Attention

Shortcomings of the model of goal-optimal attention

Model of goal-optimal attention fails along two dimensions:

- ① Attentive households' forecasts deviate more strongly from expert forecasts (though their nowcasts are more accurate).
- ② Disagreement in forecasts exceeds disagreement in nowcasts.

⇒ suggesting that agents **may not rationally update their expectations**.

⇒ propose an extension to the theory focusing on **how agents go from nowcasts to forecasts** while keeping the simple structure of the model.

Selective memory recall

Recent literature on selective memory: **current context can cue agents to retrieve memories of similar past episodes.**

(Bordalo et al., 2020, 2025; Gennaioli et al., 2024; Kahana, 2012)

⇒ Extend model by allowing **selective memory recall** to affect how agents extrapolate from nowcasts to forecasts.

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⇒ Extend model by allowing selective memory recall to affect how agents extrapolate from nowcasts to forecasts.

Key idea:

- Attentive agents observe currently high inflation.
 - They retrieve memories of past inflationary episodes.
- ⇒ They increase their perceived persistence because inflation persistence is high when inflation is high.

(Benati, 2008; Gallegos, 2023)

A model of non-goal-optimal attention

- Selective memory recall affects agent's perceived persistence:

$$\hat{\rho}_{\pi} = \rho_{\pi}(1 + \theta),$$

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- $\mathcal{I}_t$: information acquired in current period t , and D_t : agent's whole database.
- The agent recalls past experiences based on their similarity to $\mathcal{I}_t$:

$$\theta \propto S(\rho_{\pi_{t-s}}, \mathcal{I}_t, D_t)$$

- $S(\cdot)$: similarity function that overweights $\rho_{\pi_{t-s}}$ when current context $\mathcal{I}_t$ is similar to episodes s periods ago.

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- $S(\cdot)$: similarity function that overweights $\rho_{\pi_{t-s}}$ when current context $\mathcal{I}_t$ is similar to episodes s periods ago.
- ⇒ as $\rho_{\pi_{t-s}}$ is higher in periods of high inflation, agents with experiences of high inflation will have $\hat{\rho}_{\pi} > \rho_{\pi}$ when they **are attentive and observe current inflation to be high.**

A model of non-goal-optimal attention, Continued

The model overcomes the shortcomings of the goal-optimal attention model:

- **Attention becomes non-goal-optimal:** attentive agents increase expectations more than under rational expectations when inflation is high.
- Forecast disagreement increases relative to nowcast disagreement due to heterogeneity in memory databases.

A model of non-goal-optimal attention, Continued

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We cannot test the functional form of θ or $S(\cdot)$, but we can test the role of experiences:

- Agents that experienced inflation in the past have higher inflation expectations.
- Agents that experienced inflation in the past deviate more from expert forecasts.
- **Attention feedback: increase attention due to selective memory recall.** [▶ More](#)

Empirical evidence on experiences, attention and beliefs

We consider two types of experiences:

- ① experienced income or wealth loss due to inflation (elicited before inflation surge).
 - ② have lived through the 1970s oil crises.
- ⇒ interact experience measures with shock period to estimate dynamic relationship.

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Experiences and attention:

$$\begin{aligned} \text{Attention to inflation}_{it} = & \alpha \text{Prior inflation experience}_i \times \text{Shock periods}_t \\ & + \mathbf{X}'_{it}\boldsymbol{\Pi} + \phi_t + \phi_i + \epsilon_{it} \end{aligned}$$

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Differential belief responses: Households

	Attention to inflation		Expected inflation next 12 months		Absolute deviation from expert forecast	
	(1)	(2)	(3)	(4)	(5)	(6)
Inflation experience						
× 1($t \in \{21m9, 21m12\}$)	0.036*** (0.009)	0.026*** (0.009)	0.565*** (0.130)	0.185 (0.136)	0.556*** (0.127)	0.200 (0.133)
× 1($t \in \{22m3, 22m6, 22m9\}$)	0.030** (0.013)	0.054*** (0.014)	1.029*** (0.162)	0.591*** (0.175)	0.946*** (0.157)	0.563*** (0.171)
× 1($t \in \{22m12, 23m3\}$)	0.004 (0.016)	0.051*** (0.018)	1.042*** (0.188)	0.565*** (0.207)	0.934*** (0.181)	0.491** (0.200)
Experience measure	Oil crises	Past losses	Oil crises	Past losses	Oil crises	Past losses
Distinct respondents	7,126	4,913	7,925	5,404	7,925	5,404
Observations	31,347	23,820	36,451	27,913	36,451	27,913
R-squared	0.45	0.43	0.67	0.65	0.65	0.63
Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual FE	Yes	Yes	Yes	Yes	Yes	Yes

- Prior experiences associated with ...
 - stronger increases in attention to inflation in response to the shock.
 - stronger updating of inflation expectations.
 - stronger (upward) deviations from expert benchmarks.

Summary

- New panel data on HHs' and firms' attention allocation using open-ended questions.
- **Consistent with theories of goal-optimal attention:**
 - Attention varies systematically with exposure, information costs, volatility.
 - Attention associated with more frequent expectation adjustment, higher confidence, smaller misperceptions of realized inflation.
- **At odds with goal-optimal attention:**
 - Attentive agents deviate more strongly upward from expert forecasts and disagree more in forecasts than in nowcasts.
- **Extended model with selective memory recall:**
 - Reconciles theory and empirics, important role of experiences: confirmed in the data.
 - **Interference of other variables:** paying attention to energy prices also increases inflation expectations, consistent with the model. [▶ More](#)
- **Implications:**
 - Macro modeling: Examine aggregate implications of selective memory recall.
 - Use of open-ended data to measure economic attention in other contexts.

Appendix

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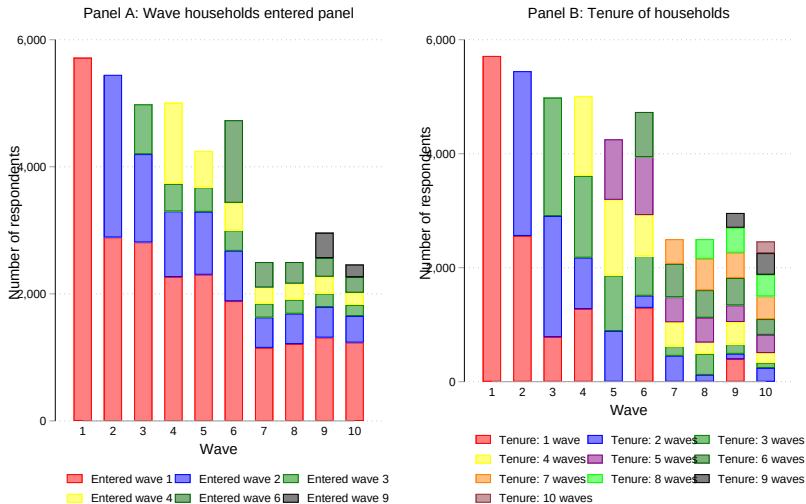
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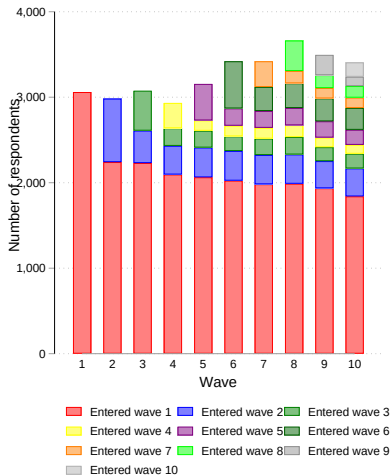
Survey responses across waves: Households

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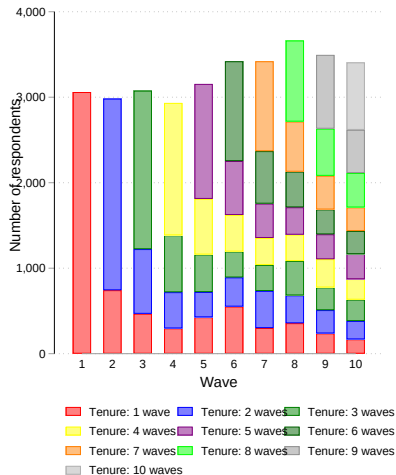
Survey responses across waves: Firms

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Panel C: Wave firms entered panel

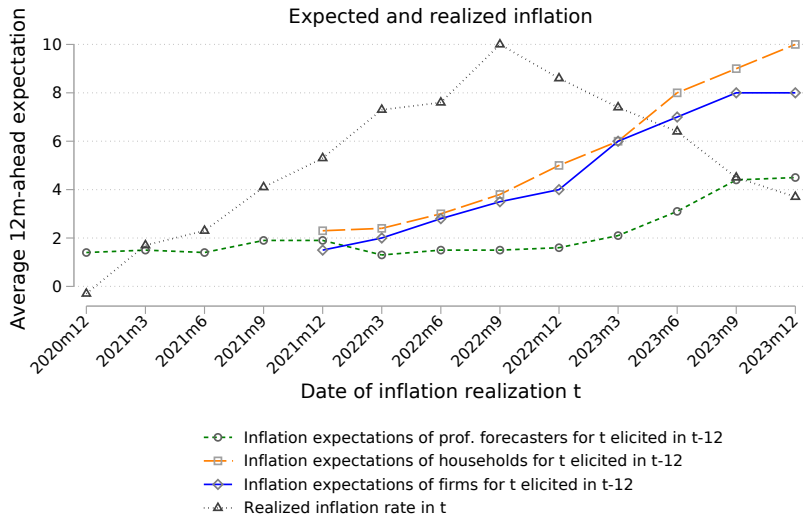


Panel D: Tenure of firms



	GSOEP	Survey samples					
	(1) Mean	(2) Mean	(3) p25	(4) Median	(5) p75	(6) SD	(7) N
Panel A: Households							
Female	0.51	0.45	0.00	0.00	1.00	0.50	40,552
Age	51.19	52.53	40.00	50.00	60.00	13.85	40,552
East	0.17	0.17	0.00	0.00	0.00	0.38	40,552
Log(HH net income)	7.96	7.78	7.60	8.01	8.36	0.69	40,552
At least highschool	0.39	0.50	0.00	1.00	1.00	0.50	40,552
Employed	0.64	0.59	0.00	1.00	1.00	0.49	38,421
Homeowner	0.49	0.48	0.00	0.00	1.00	0.50	40,552
Stockowner	0.26	0.42	0.00	0.00	1.00	0.49	40,552
Panel B: Firms							
Employees		326.00	14.00	40.00	125.00	2336.81	32,539
Export share		0.15	0.00	0.01	0.24	0.24	17,101
Manufacturing firm		0.29	0.00	0.00	1.00	0.45	32,612
Services firm		0.41	0.00	0.00	1.00	0.49	32,612
Construction firm		0.08	0.00	0.00	0.00	0.27	32,612
Retail/wholesale firm		0.22	0.00	0.00	0.00	0.41	32,612
High influence on decisions in firm		0.78	1.00	1.00	1.00	0.42	20,417

Unexpected shock to inflation

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Hand-coding vs. word count for topic “inflation”

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	Hand-coded	Automated word count					Correlation
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
		Inflation	Price	Cost	Expensive	Joint word count	hand-coded vs. joint word count
Panel A: Households							
Wave 1: 2020m12	0.03	0.01	0.02	0.01	0.03	0.05	0.60
Wave 2: 2021m3	0.04	0.01	0.02	0.01	0.03	0.06	0.75
Wave 3: 2021m6	0.07	0.02	0.04	0.02	0.04	0.10	0.81
Wave 4: 2021m9	0.09	0.04	0.05	0.02	0.05	0.13	0.78
Wave 5: 2021m12	0.16	0.07	0.07	0.02	0.04	0.17	0.88
Wave 6: 2022m3	0.28	0.09	0.14	0.04	0.06	0.27	0.88
Wave 7: 2022m6	0.32	0.21	0.17	0.05	0.06	0.39	0.82
Wave 8: 2022m9	0.38	0.20	0.20	0.08	0.06	0.43	0.86
Wave 9: 2022m12	0.33	0.23	0.19	0.06	0.07	0.42	0.80
Wave 10: 2023m3	0.35	0.23	0.18	0.06	0.08	0.44	0.82
Total (Waves 1-10)	0.19	0.09	0.09	0.03	0.05	0.22	0.84
Panel B: Firms							
Wave 1: 2020m12	0.05	0.01	0.04	0.01	0.03	0.09	0.69
Wave 2: 2021m3	0.10	0.01	0.07	0.01	0.04	0.14	0.79
Wave 3: 2021m6	0.19	0.02	0.15	0.03	0.03	0.23	0.87
Wave 4: 2021m9	0.19	0.03	0.14	0.04	0.06	0.28	0.78
Wave 5: 2021m12	0.25	0.07	0.16	0.04	0.02	0.28	0.89
Wave 6: 2022m3	0.33	0.09	0.24	0.07	0.02	0.39	0.76
Wave 7: 2022m6	0.43	0.19	0.24	0.07	0.03	0.48	0.82
Wave 8: 2022m9	0.42	0.19	0.28	0.10	0.02	0.52	0.75
Wave 9: 2022m12	0.40	0.20	0.22	0.09	0.02	0.46	0.76
Wave 10: 2023m3	0.35	0.20	0.16	0.06	0.02	0.41	0.79
Total (Waves 1-10)	0.28	0.11	0.16	0.06	0.03	0.34	0.81

Hand-coding vs. AI-based coding

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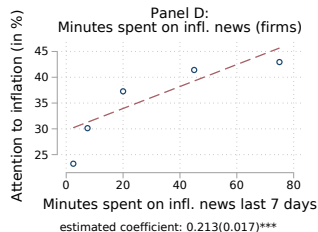
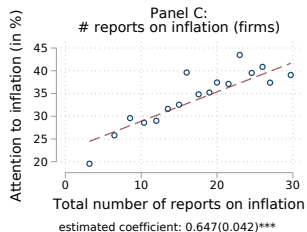
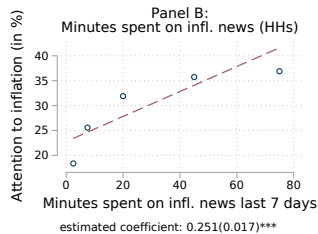
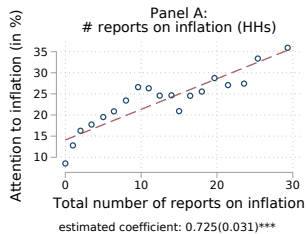
	Hand-coded				
	(1) Covid-19	(2) Inflation	(3) Growth	(4) Any macro	(5) Any personal
AI-coded: Covid-19	0.997*** (0.004)	-0.079 (0.070)	-0.004 (0.007)		
AI-coded: Inflation	-0.006 (0.006)	0.808*** (0.032)	0.015 (0.013)		
AI-coded: Growth	-0.003 (0.004)	0.421** (0.205)	0.746*** (0.219)		
AI-coded: Any macro topic				0.727*** (0.051)	0.014 (0.045)
AI-coded: Any household-level topic				0.004 (0.050)	0.680*** (0.058)
Observations	200	200	200	200	200
R-squared	0.66	0.52	0.75	0.53	0.52
Mean dep. var.	0.01	0.36	0.01	0.45	0.72

Validation: Structured question on attention

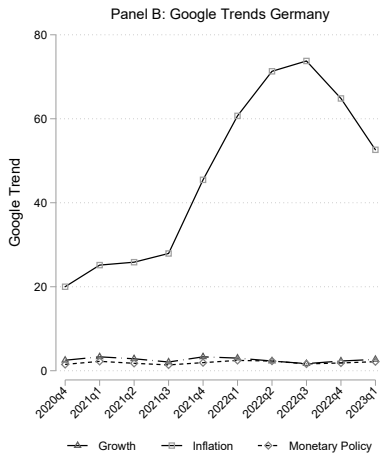
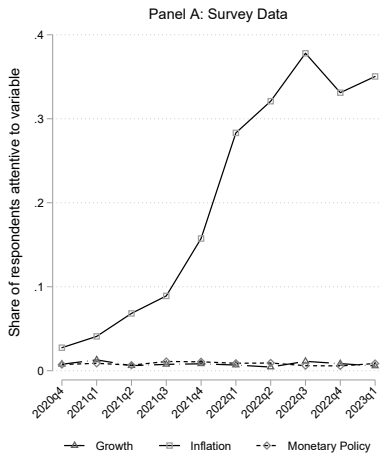
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	Open-ended					
	(1)	(2)	(3)	(4)	(5)	(6)
	Covid-19	Inflation	Monetary policy	Growth	Any macro topic	Any household-level topic
Structured: Covid-19	0.098* (0.053)	-0.032 (0.086)	-0.012* (0.007)	0.012 (0.040)		
Structured: Inflation	0.008* (0.005)	0.159*** (0.041)	0.008* (0.004)	0.002 (0.014)		
Structured: Monetary policy	-0.008 (0.005)	0.040 (0.059)	0.032 (0.024)	0.039* (0.023)		
Structured: Growth	-0.018* (0.010)	0.089 (0.062)	-0.006 (0.020)	0.072** (0.029)		
Structured: Any macro topic					0.151*** (0.049)	-0.032 (0.050)
Structured: Any household-level topic					-0.072 (0.203)	0.469** (0.192)
Observations	468	468	468	468	468	468
R-squared	0.10	0.04	0.02	0.04	0.01	0.02
Mean dep. var.	0.01	0.26	0.01	0.03	0.29	0.79

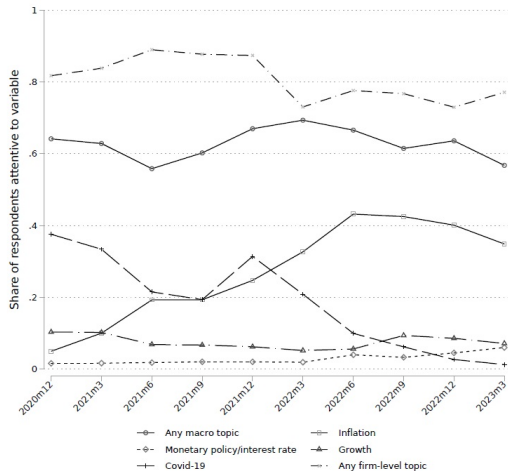
Validation: Attention and news consumption

[▶ Back](#)

Validation: Attention and Google searches

[▶ Back](#)

Attention allocation over time: Firms

[▶ Back](#)

- Attention shifts from Covid-19 to inflation over sample period.

Variance decomposition of attention

	Panel A: Households R2 (%) of panel regression			(4) Obs.	Panel B: Firms R2 (%) of panel regression			(8) Obs.
	(1)	(2)	(3)		(5)	(6)	(7)	
	Indiv. FE	Time FE	Time FE + Indiv. FE		Indiv. FE	Time FE	Time FE + Indiv. FE	
<i>Any macro topic</i>	41.1	3.2	43.3	31,348	33.0	0.7	33.7	27,554
Inflation	38.1	10.1	44.9	31,348	31.8	8.0	38.7	27,554
Monetary policy	27.9	0.0	28.0	31,348	34.3	0.7	35.0	27,554
Growth	25.2	0.1	25.3	31,348	27.4	0.5	27.8	27,554
Covid-19	37.9	2.7	39.6	31,348	32.2	10.5	41.1	27,554
<i>Any household-/firm-level topic</i>	42.3	1.4	43.3	31,348	32.2	2.0	33.7	27,554

- Theory predicts systematic variation in attention.
- Individual fixed effects are an important source of variation.
- Time fixed effects explain a meaningful but smaller part of the variation. [▶ back](#)

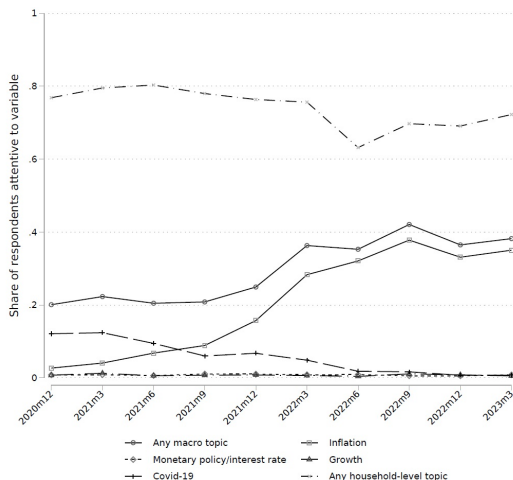
Cross-sectional variation in attention: Firms [▶ Back](#)

	Attention			
	(1) Inflation	(2) Monetary policy	(3) Growth	(4) Any macro topic
High perceived general exposure	0.101*** (0.007)	0.034*** (0.003)	0.026*** (0.005)	0.081*** (0.009)
High influence on decisions in firm	-0.035*** (0.013)	-0.002 (0.005)	-0.007 (0.007)	0.001 (0.014)
Log(Employees)	0.007** (0.003)	0.002* (0.001)	0.003** (0.002)	0.015*** (0.003)
Export share	-0.050** (0.024)	-0.009 (0.007)	0.027* (0.015)	0.013 (0.028)
Services firm	-0.122*** (0.010)	0.027*** (0.004)	-0.002 (0.006)	-0.014 (0.012)
Construction firm	-0.005 (0.017)	0.059*** (0.010)	0.001 (0.010)	-0.001 (0.018)
Retail/Wholesale firm	-0.051*** (0.012)	0.020*** (0.004)	-0.014** (0.007)	0.002 (0.013)
Observations	6,051	6,047	6,047	6,028
R-squared	0.08	0.04	0.01	0.02

- Consistent with model predictions, attention increases with proxies for perceived exposure to the variable to be forecast.

Time variation in attention: Households

► Firms



- Attention shifts from Covid-19 to inflation, which, through the lens of the model, might reflect higher inflation volatility and/or lower information costs due to more media coverage. [► back](#)

Disagreement

	Households			Firms		
	(1)	(2)	(3)	(4)	(5)	(6)
	SD	IQR	p90-p10	SD	IQR	p90-p10
Expected inflation						
Attentive to inflation (a)	4.86	3.08	8.00	2.71	2.42	5.12
Inattentive to inflation (b)	5.97	2.98	8.80	2.60	2.17	4.77
Perceived current inflation						
Attentive to inflation (c)	4.06	2.66	5.16			
Inattentive to inflation (d)	5.25	2.80	6.82			
p-value: (a)=(c)	0.00	0.00	0.00			

- Disagreement among firms lower than among households
- Attentive households disagree less than inattentive households (opposite for firms)
- Disagreement in forecasts *exceeds* disagreement in nowcasts, at odds with the model

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⇒ Agent pays more attention to inflation due to selective memory recall:

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⇒ selective memory recall predicts important role for experiences for belief formation and attention allocation... even among agents with same B, σ_π^2 and κ [▶ back](#)

Interference of attention to other variables

Extension: Attention to other variables may interfere through selective memory recall (Bordalo et al., 2025).

- Attention to energy prices and observation of high current energy prices cues recall of past episodes of high energy prices.
- When forecasting inflation, the agent thinks of inflation in past episodes of high energy prices $\Rightarrow$ higher inflation expectations.

Formally (x : energy prices):

$$\frac{\partial \theta}{\partial \text{cov}(\rho_{\pi_{t-s}}, (x_{t-s} - x_{t|t})^2)} < 0$$

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We confirm these predictions in the data.